

Merck Pharma and Covid Vaccines

A case for decision making under uncertainty

Background

In September 2020 Dr. Patricia Okuna, Director of the research and development (R&D) unit at Merck Pharmaceuticals, had to make a decision: should they move forward with the current version ("version A") of the Coronavirus vaccine they developed, or should they wait another three months, which is the time it would take to develop one more version of their vaccine ("version B"). Waiting means that they risk becoming less relevant as their competitors make progress, while going out for testing now risks a lower chance of a successful vaccine.

Developing Vaccines

For more than 125 years, Merck (known as MSD outside North America) has been bringing forward medicines and vaccines for many of the world's most challenging diseases in pursuit of their mission to save and improve lives. With over seventy-one thousand employees across the world, Merck has almost sixteen thousand individuals engaged in various aspects of R&D and the company invested almost ten billion dollars in R&D in 2019.

Though not their largest facility, the South San Francisco center that was completed in 2019 houses over 300 Scientists under Dr. Okuna's leadership. The state-of-the-art laboratories support cutting-edge research to develop medications and vaccines. The method pursued to develop a vaccine for the Coronavirus is to weaken the virus so that it reproduces very poorly once inside the body, similar to influenza vaccines that are made this way. Natural viruses typically reproduce themselves thousands of times in the body, causing disease and even death. Vaccine viruses developed in labs typically reproduce fewer than 20 times, meaning that they don't cause disease, but they are able to cause the body to produce antibodies against infection in the future. This method requires only one or two doses to provide life-long immunity, but are usually not safe for people with weakened immune systems

The Decision Problem

Dr. Okuna had to make the decision of moving forward with version A or to delay launching their clinical by 3 months in the hope of having a highly effective version B. From their previous R&D experience, Dr. Okuna concluded that there was a fifty-fifty chance that waiting 3 months would result in an effective version B of their vaccine. If such an effective version B did materialize, Dr. Okuna and her team estimated that there was a 40% chance that it would have a **high efficacy** rate and a 60% chance that it would have a **mediocre efficacy** rate. A high efficacy vaccine would yield profits of \$5.60 per vaccine. A mediocre efficacy rate would result in profits of about \$2 per vaccine, which is about \$0.85 less than what Dr. Okuna believes they could obtain by **proceeding immediately with version A**. Merck always has the option of licensing a mediocre version B to a generic producer. This option would bring only half as much profit, but would avoid damaging Merck's reputation, which may be at risk by selling inferior product.

If Dr. Okuna decides to wait and attempt to develop version B, but version B does not successfully materialize, the delayed 3 months would still give their team time to make small improvements to version A and there are three possible scenarios. With luck, version A would have a **very good efficacy and there would be weak competition**, yielding a profit of \$3.50 per vaccine. With less luck, **strong competition** would cause profits to be only \$3.00 per vaccine. Dr. Okuna and her team estimated that these two scenarios were equally likely. A third scenario from waiting and not having version B materialize is that version A's efficacy is **good but not very good**, in which case profits would be \$2.50 per vaccine regardless of competition. Dr. Okuna and her team estimated that that this event had only about 0.2 probability of occurring.

After taking all these facts into account Dr. Okuna was struggling with the decision, but their estimates did paint a relatively clear picture of how to proceed.